Question 7

I would use Kruskal's algorithm which list all roads and sort them from smallest weight to largest and take the smallest road and add it to the tree only if it does not form a cycle. This is repeated until $N-1 = 19$ roads are selected.

Reasons for choosing Kruskal's

1. ~~The~~ The graph is sparse : 20 nodes and 30 roads, while a complete graph of 20 nodes would have ~~190~~ more Kruskal's running time, $O(e \log e)$, depends on the number of edges, so it works well.

2. The graph is stored as a list of edges, which is exactly what Kruskal's needs ~~(sort the list, then~~

3. It is simple to implement, using only sorting and a ~~cycle~~ cycle check.

4. It can connect disconnected graph : If fewer than 19 edges are selected, some locations are not connected

* On the other hand, Prim's algorithm grows one tree from a starting node and works best on dense graphs

Therefore, ~~Both~~ Kruskal's is the better fit for a small, sparse graph stored as an edge list like ~~this~~ This.